4.3 Hypothesis tests using bootstrap

For hypothesis testing it is important that the **empirical sampling distribution is generated under the null** H_0.

This means that usually **parametric bootstrap should be used!**

(Or data rescaling if suitable –see later.)

Example 4.1 (cont.) – ***Parametric bootstrap***:
We will use bootstrap to test

H_0: the true mean failure time (say μ) is 140 hours
vs H_1: the true mean failure time is *less* than 140 hours

Now the empirical sampling distribution (and hence the bootstrap samples) must come from a distribution with $\mu = 140$.

We therefore **assume** that data, under H_0, come from Exp(1/140).

We first obtain an estimate of μ ($\hat{\mu} = 95.7$)
then obtain B bootstrap values $\hat{\mu}_1^*, \ldots, \hat{\mu}_B^*$
and finally compute p-value as

$$p = \frac{\sum_{j=1}^{B} I(\hat{\mu}_j^* \leq \hat{\mu}) + 1}{B + 1}$$

, where $I(\cdot)$ is the indicator function.

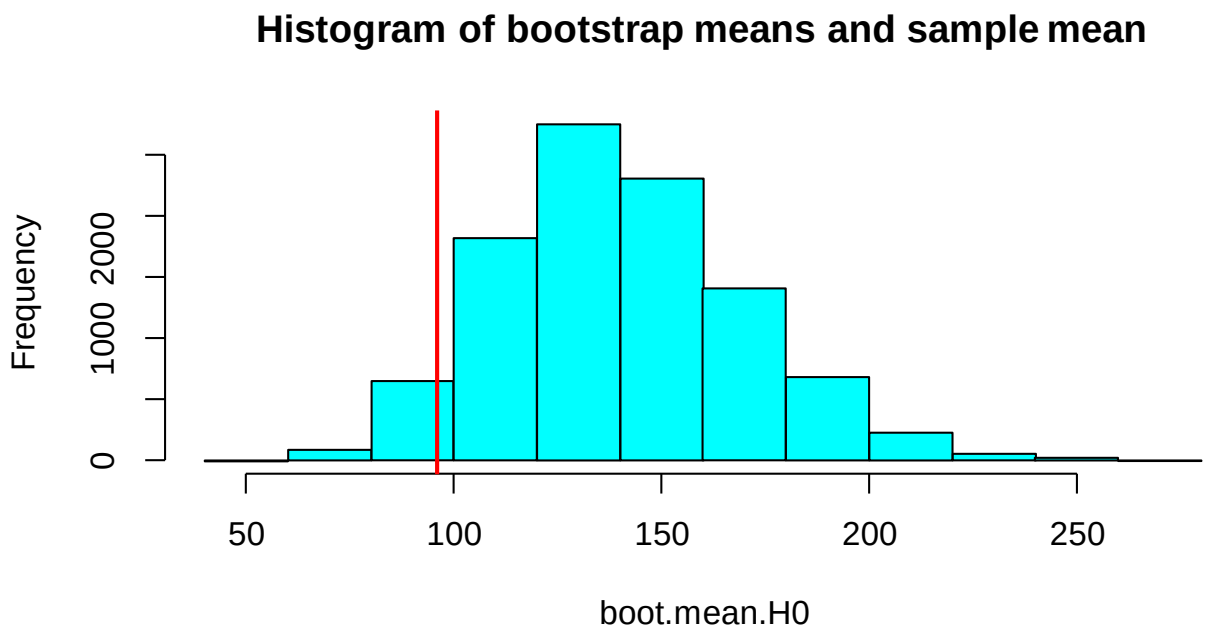
In R this can be done as follows:

```
###  Perform test for mean = mean.hyp v mean < mean.hyp  (H1: μ < 140)
###  Parametric bootstrap is well suited for hypothesis testing

  B = 10000
  mean.hyp = 140
  boot.mean.H0 = numeric(B)     # Create vector for bootstrap values
  for (i in 1: B) {
      y.boot = rexp (23, rate = (1/mean.hyp))   # Notice this is under H0
      boot.mean.H0[i] = mean (y.boot)
  }

###  Obtain sampling distribution under H0 and p-value
  hist(boot.mean.H0, col="cyan", main="Histogram of bootstrap means and sample mean")
  mu.hat = mean(failures)
  abline (v = mu.hat, col="red", lwd=2.0)
  boot.pval = (1 + length( boot.mean.H0[boot.mean.H0 <= mu.hat]) ) / (B+1)
  boot.pval
```

This code will give



and p-value = 0.049

Note that a *test based on asymptotic normality* will give

t.test (failures,mu=140, alternative="less")

... t = -1.7812, df = 22, p-value = 0.04435

Example 4.1 (cont.) – ***Non-parametric bootstrap***:

Again, for failures data, test

H_0: the true mean failure time (say μ) is 140 hours

vs H_1: the true mean failure time is *less* than 140 hours

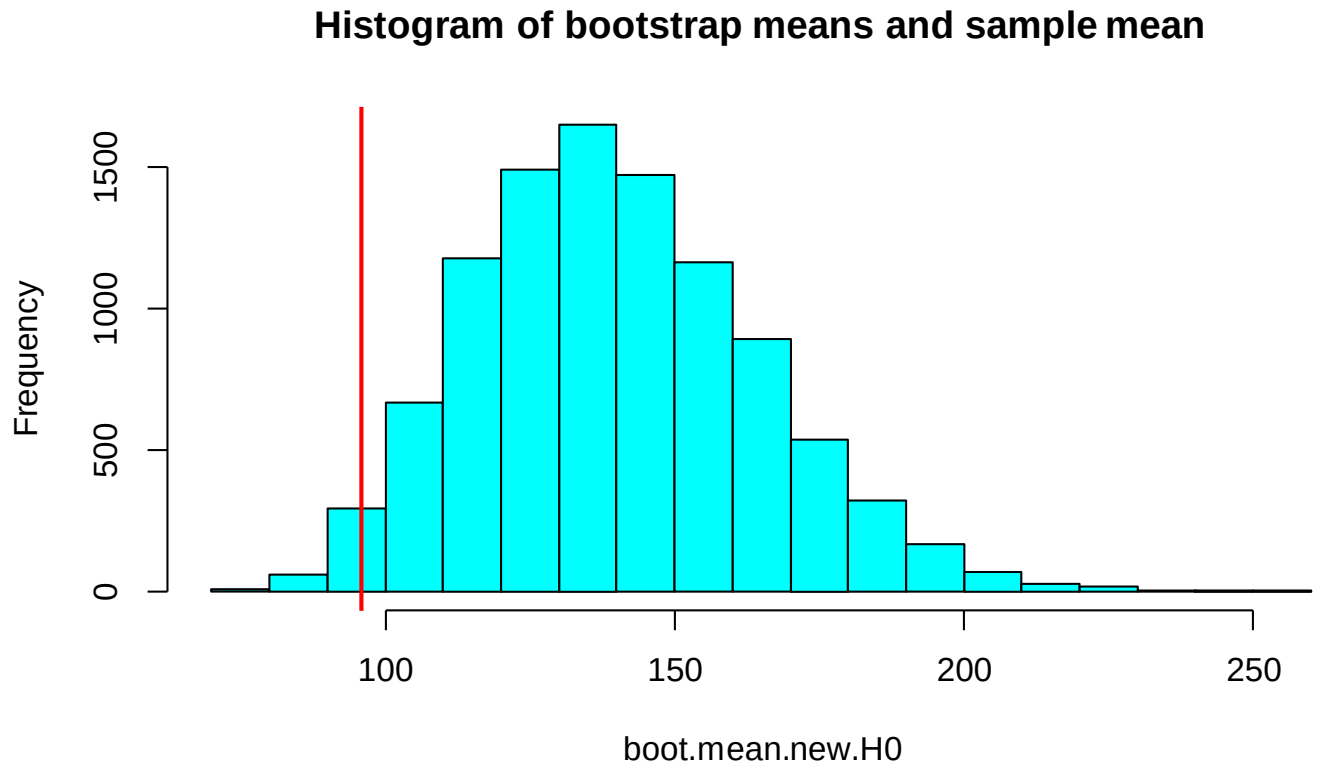
- First rescale data so that H_0 is true:
  $ynew_i = y_i - \bar{y} + 140$
- Then resample new data (with replacement) and proceed with bootstrap test as before.
- Note that suitable rescaling not always straightforward!

R Code:

```
# Rescale data so that H0 is true
failures.new = failures - mean(failures) + mean.hyp
mean (failures.new)

B=10000
boot.mean.new.H0 = numeric(B)    #Create vector for bootstrap values
for (i in 1: B) {
        boot.new.y = sample (failures.new,23,replace=T)     #This is under H0
        boot.mean.new.H0[i] = mean(boot.new.y)
}
hist(boot.mean.H0, col="cyan", main="Histogram of bootstrap means and sample mean")
mu.hat = mean(failures)
abline (v = mu.hat, col="red", lwd=2.0)
boot.pval = (1 + length( boot.mean.H0[boot.mean.H0 <= mu.hat]) ) / (B+1) ; boot.pval
```

Then sampling distribution and p-value obtained as before:



- This is the empirical distribution of the sample mean from a population with mean 140.
- Notice that it is not symmetric - bootstrap justified!
- Observed sample mean = 95.7. How likely is such an extreme (small) value, under H_0?

p-value = 0.02